# MILITARY SPECIFICATION

# SURVEYING SET: TRIANGULATION RECONNAISSANCE, SPECIALIZED; PACKAGING OF

## 1. SCOPE

1.1 This specification covers the requirements for preservation, packaging, and packing of a specialized, triangulation reconnaissance, surveying set, FSN 6675-641-3603 (No. SC 6675-93-CL-E26).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-636 | - Box, Fiberboard. |
| PPP-B-640 | - Boxes, Fiberboard, Corrugated, Triple-Wall. |

Military

| | |
|---|---|
| MIL-P-116 | - Preservation, Methods of. |

STANDARDS

Federal

| | |
|---|---|
| FED. TEST METHOD STD. NO. 101 | - Preservation, Packaging, and Packing Materials: Test Procedures. |

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |



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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and in applicable specifications. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2 Preproduction pack. The preproduction pack is defined as a production pack which will be identical to the pack which the supplier will furnish in fulfillment of the contract. Before a number of sets is packed, a preproduction pack shall be made, examined, and subjected to the tests specified for the applicable level to determine conformance to the preservation, packaging, packing, and marking requirements. Approval of the preproduction pack shall not relieve the supplier of his obligation to preserve, package, pack, and mark the sets in accordance with this specification. Any changes or deviations of production packs from the approved preproduction pack shall be subject to the approval of the contracting officer.

3.3 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

3.3.1 Level A.

3.3.1.1 Preservation. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116. Methods of preservation indicated by symbols herein shall conform to the applicable symbols listed in MIL-P-116.

3.3.1.2 Altimeter barometers. Each altimeter barometer shall be placed in the carrying case provided and individually preserved method IIb. The carrying case shall be the inner container and the outer container shall conform to PPP-B-636, W5c, style optional. The outer container shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.3 Axes. The axes shall be preserved method I, using type P-1 preservative on metal surfaces.



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3.3.1.4 Ax sheaths, map and photograph case, and pen and pencil sheaths. Each item shall be individually preserved method IC-1.

3.3.1.5 Water bags. Preservation shall not be required.

3.3.1.6 Safety belt. The belt shall be preserved method III, using a container conforming to PPP-B-636, W5c, style optional. The container shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.7 Binoculars. Each binocular, with eyepiece and aperture lens covers in place, shall be placed in the carrying case and preserved method IIb. The carrying case shall be the inner container and the outer container shall conform to PPP-B-636, W5c, style optional. The outer container shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.8 Climber's set. Metal surfaces of the climber's set requiring the application of a contact preservative shall be coated with type P-1 preservative. The climber's set shall be packaged in a box conforming to PPP-B-636, W5c, style optional. The box shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.9 Compasses. Each compass shall be placed in the carrying case provided and individually preserved method IC-1.

3.3.1.10 Surveying levels. Each level shall be placed in the carrying case provided and individually preserved method IC-1.

3.3.1.11 Machetes. Each machete shall be preserved method I, applying type P-1 preservative to the blade.

3.3.1.12 Pencils, protractors, drafting scales, and drafting triangles. Each item contained in a vendor's carton shall be preserved method IC-1.

3.3.1.13 Plane table. The plane-table board shall be placed in the case provided and preserved method III, using a box conforming to PPP-B-636, W5c, style optional. The box shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.1.14 Tripod. The tripod shall be collapsed to the minimum dimension, placed in the carrying case provided, and preserved method III, using a box conforming to PPP-B-636, W5c, style optional. The box shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.



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3.3.1.15 Machete sheaths. Preservation shall not be required.

3.3.1.16 Safety strap. The strap shall be preserved method IC-1.

3.3.1.17 Measuring tape. The tape shall be preserved method IC-1, applying type P-7 preservative to unprotected metal surfaces.

3.3.1.18 Mailing and filing tubes. The tubes shall be individually preserved method IC-1.

3.3.2 Level B. Each component of the triangulation surveying set, with the exception of the altimeter barometers and binoculars, shall be preserved as specified for level A.

3.3.2.1 Altimeter barometers. Each altimeter barometer shall be placed in the carrying case provided and individually preserved method IA-14. The carrying case shall be the inner container and the outer container shall conform to PPP-B-636, W5c, style optional. The outer container shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.2.2 Binoculars. Each binocular, with eyepiece and aperture lens covers in place, shall be placed in the carrying case and preserved method IA-14. The carrying case shall be the inner container and the outer container shall conform to PPP-B-636, W5c, style optional. The outer container shall be closed and waterproof sealed with tape in accordance with the appendix to the box specification.

3.3.3 Level C. The components of the triangulation surveying set shall be preserved and packaged to afford protection against deterioration and damage during shipment from the supplier to the initial destination for immediate use.

3.4 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.4.1 Level A. The components of the triangulation surveying set shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. The box shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

3.4.2 Level B. The components of the triangulation surveying set shall be packed together in a close-fitting box conforming to PPP-B-601, domestic type, style optional, or PPP-B-640, class 2, style A, B, C, or D. The containers shall be closed and strapped in accordance with the appendix to



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the container specification. The strapping shall not be required to be zinc coated.

3.4.3 Level C. The components of the triangulation surveying set shall be packed to insure carrier acceptance and safe delivery to destination at the lowest rates. Shipping containers, as a minimum, shall comply in all respects with common carrier rules and regulations applicable to the mode of transportation.

3.5 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

3.6 Workmanship. Preservation, packaging, packing, and marking shall be accomplished in a manner that will assure protection and identification provided by this specification and referenced documents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction pack inspection.
(b) Quality conformance inspection.

4.3 Preproduction pack inspection.

4.3.1 Examination. The pack shall be examined for the defects specified in 4.4.3. Presence of one or more defects shall be cause for rejection.



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4.3.2 Tests. The preproduction pack for levels A and B shall be subjected to the free-fall drop test in accordance with FED. TEST METHOD STD. NO. 101, method 216. Rupture or deformation of the pack or packages, or damage to the components of the surveying set shall constitute failure of the test and shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Unit of product. For the purpose of inspection, a completed pack of any one level prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

101. Materials not as specified for level A or B. Each incorrect material shall constitute one defect.
102. Methods not as specified for level A or B. Each incorrect method shall constitute one defect.
103. Containers not as specified for level A or B. Each incorrect container shall constitute one defect.
104. Marking illegible, incomplete or incorrect for level A, B, or C. Each incorrect, incomplete, or illegible marking shall constitute one defect.

5. PREPARATION FOR DELIVERY

(Not applicable to this specification)

6. NOTES

6.1 Intended use. This specification is intended to be used for direct reference in contracts or orders to cover the preservation, packaging, packing, and marking of surveying set: triangulation reconnaissance, specialized; FSN 6675-641-3603 (SC 6675-93-CL-E26).



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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging and level of packing required (see 3.3 and 3.4). Level B preservation and packaging is intended to provide limited but economical protection and should be specified only when it is determined the surveying sets will be held in covered storage.

Custodians:

Army - ME

Preparing activity:

Army - ME

Project No. 6675-A104



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-S-52541(ME) - Surveying Set: Triangulation Reconnaissance, Specialized Packaging of

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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